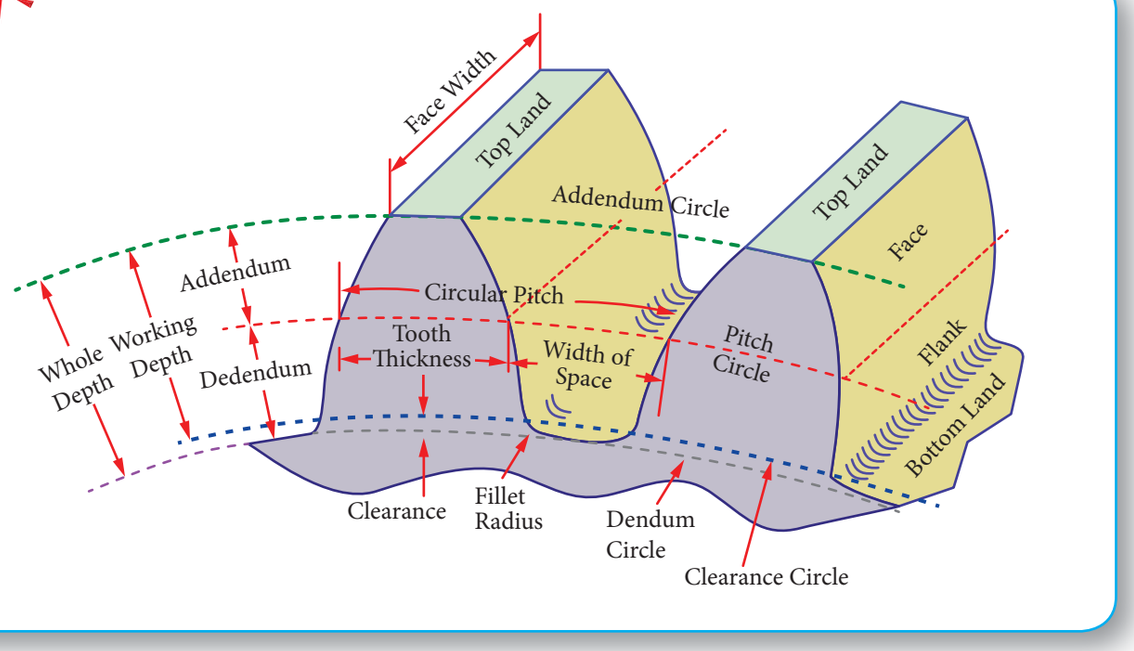
DO YOU KNOW?




## Belt drive advantages and disadvantages

### Advantages

1. Absorbs noise and vibrations
2. Protects from overload
3. Needs little maintenance
4. Allows misalignment (Parallel shafts)

### Disadvantages

1. Speed ratio is not constant (Slip & Stretch)
2. Speed limited- 2000 m/min
3. Endless belts needs special attention to install
4. Belt may slip from Pulley while rotate

## 9.5 GEARS

Gears are used to transmit power between rotating parts to operate various machines. The power transmission is achieved without any slip. It is also advantageous in the sense that higher velocity ratio can be achieved in limited space. Only parallel shafts are connected by belt drive whereas parallel non-parallel and perpendicular shafts are connected by means of gears to transmit power.

## FORMS OF GEARS

There are different forms of gears namely:

1. Spur gear.
2. Helical gear.
3. Bevel gear.
4. Rack and pinion gear.
5. Worm and worm gear.

## 1. SPUR GEAR

Spur gears have their teeth elements parallel to the rotating shafts. These gears are used to transmit power between parallel shafts. A small sized gear is called pinion.



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